171773: quartz sandstone, Ingebong Hills

Location and sampling

NABBERU (SG 51-5), RHODES (3147)
MGA Zone 51, 312250E 7208470N

Sampled on 9 June 2002

The sample was taken from close to the base of a sandstone bluff that forms Ingebong Hills, northeast of the parking area off the Canning Stock Route.

Tectonic unit/relations

The sample is from near the base of the Wonyulgunna Sandstone, the basal unit of the Mesoproterozoic succession on RHODES (3147) and METHWIN (3047), within a facies showing very large scale cross-bedding (Hocking et al., 2003). The Wonyulgunna Sandstone was previously considered to be part of the Collier Group (Hocking and Jones, 2002; Martin and Thorne, 2002) but is now considered to be part of the Scorpion Group (Hocking et al., 2003), which is a probable correlative of the Edmund Group, Edmund Basin. Palaeocurrent measurements indicate derivation broadly from the south to west. The sample was collected to constrain the provenance, possible correlatives, and depositional age of the Wonyulgunna Sandstone.

Petrographic description

The sample consists principally of subrounded to rounded and elliptical single-crystal quartz grains (85 vol.%) and optically continuous silica overgrowths (15 vol.%), with trace amounts of accessory sericite, tourmaline, and zircon. This is a coarse-grained, quartz-rich sandstone with fairly common quartz overgrowths, minor lithic grains, and incipiently stylolitic inter-granular sericite. Subrounded to rounded and elliptical single-crystal quartz grains dominate, with a maximum grain size between 0.8 and 1 mm. Optically continuous overgrowths comprise about 15% of the thin section area. About 2 vol.% of the quartz grain aggregate is composed of polycrystalline quartz grains, locally clouded by limonite, clay, or both. There are rare detrital grains of slaty composition with schistose sericite and extremely fine-grained quartz. The single-crystal quartz grains are cemented by sporadic optically continuous overgrowths, and locally have deformation lamellae indicating low-grade metamorphism involving weak deformation. Bluish-green and, less commonly, orange-brown tourmaline is present, mostly as rounded grains up to 0.2 mm in diameter. Interstitial sericite is scattered through the quartz grain aggregate, and within incipient stylolitic networks and laminae, indicating a pressure solution origin rather than authigenic quartz deposition.

Zircon morphology

The zircons isolated from this sample are generally colourless or pale yellowish- to greenish-brown, between 70 × 90 µm and 250 × 350 µm in size, internally structureless or with patchy brown discolouration, and slightly elongate or equant and rounded or elliptical in shape. Fluid and mineral inclusions are common. The surfaces of most grains are pitted, consistent with detrital transport. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 1 April 2003. The counter deadtime during the analysis session was 24 ns. Twelve analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 2.55% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses.

Results

Thirty-five analyses were obtained from 35 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant and indicate a wide range of $^{207}Pb/^{206}Pb$ dates, from c. 1750 to 2700 Ma. Twenty–four concordant and highly discordant analyses of 24 zircons (3.1, 4.1, 5.1, 6.1, 7.1, 8.1, 11.1, 12.1, 13.1, 14.1, 15.1, 16.1, 17.1, 19.1, 22.1 23.1, 25.1, 27.1, 28.1, 30.1, 31.1, 33.1, 34.1, 35.1), assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 1798 ± 9 Ma (chi-squared = 0.77). The remaining analyses indicate older $^{207}Pb/^{206}Pb$ dates and cannot be confidently grouped.

Most grains with analyses belonging to Group 1 have pitted surfaces and are inferred to be detrital. Consequently, the weighted mean $^{207}Pb/^{206}Pb$ date of 1798 ± 9 Ma indicated by the 24 concordant and highly discordant analyses of Group 1 is interpreted as a maximum age for deposition of the sandstone.

Possible source rocks within the western part of Australia having ages matching those of the zircons within this sample include the Rudall (Nelson, 1995, 1996), Arunta (Williams et al., 1996 and references cited therein) and Gascoyne (Nelson, 1997, 1998, 1999, 2000, this compilation) Complexes. Grains 2.1, 18.1, 20.1, and 32.1 could have been derived from the Yilgarn or Pilbara Cratons.

Recommended reference for this publication:

NELSON, D. R., 2005, 171773: quartz sandstone, Ingebong Hills; Geochronology dataset 543; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 01/04/2003; Data released: 30/06/2005

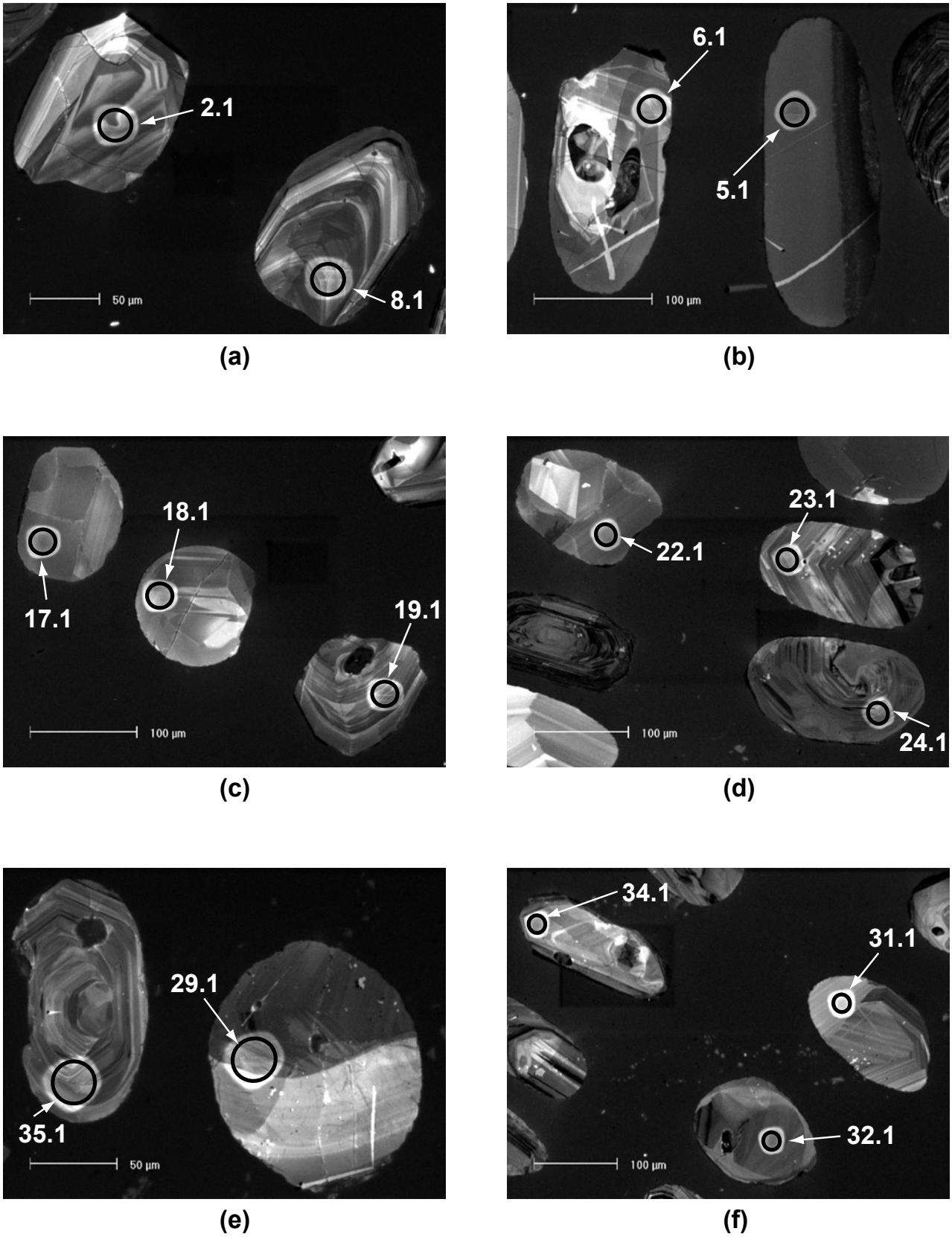

Figure 1. Cathodoluminescence images of representative zircons from sample 171773: quartz sandstone, Ingebong Hills

Table 1. Ion microprobe analytical results for zircons from sample 171773: quartz sandstone, Ingebong Hills

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	133	89	49	0.081	0.12288	0.00117	0.15752	0.00210	0.3353	0.0087	5.681	0.163	93	1 998	17
2.1	97	138	65	0.022	0.17740	0.00127	0.39539	0.00274	0.4922	0.0128	12.039	0.336	98	2 629	12
3.1	119	135	30	0.389	0.11055	0.00185	0.25310	0.00417	0.2152	0.0055	3.280	0.106	69	1 808	30
4.1	191	88	61	0.087	0.10967	0.00111	0.12747	0.00208	0.2998	0.0077	4.534	0.130	94	1 794	18
5.1	170	369	82	0.128	0.10903	0.00103	0.62175	0.00359	0.3209	0.0083	4.824	0.138	101	1 783	17
6.1	123	88	43	0.065	0.11012	0.00111	0.21048	0.00225	0.3107	0.0080	4.718	0.136	97	1 801	18
7.1	81	143	37	0.079	0.10948	0.00152	0.50402	0.00463	0.3261	0.0085	4.922	0.153	102	1 791	25
8.1	126	86	44	0.123	0.10894	0.00136	0.19649	0.00285	0.3111	0.0080	4.673	0.140	98	1 782	23
9.1	128	48	53	0.123	0.13612	0.00111	0.10795	0.00158	0.3837	0.0099	7.202	0.202	96	2 178	14
10.1	391	219	117	0.098	0.14155	0.00080	0.07490	0.00111	0.2854	0.0073	5.570	0.150	72	2 246	10
11.1	142	78	48	0.266	0.10990	0.00124	0.15648	0.00238	0.3075	0.0079	4.660	0.137	96	1 798	20
12.1	272	399	85	0.147	0.11027	0.00088	0.32143	0.00217	0.2510	0.0064	3.816	0.106	80	1 804	15
13.1	186	81	57	0.038	0.11158	0.00157	0.13254	0.00331	0.2872	0.0074	4.418	0.136	89	1 825	26
14.1	175	107	61	-0.047	0.11053	0.00173	0.17895	0.00380	0.3128	0.0081	4.766	0.151	97	1 808	28
15.1	58	48	21	-0.013	0.11084	0.00149	0.24082	0.00314	0.3075	0.0081	4.700	0.145	95	1 813	24
16.1	136	89	47	0.167	0.11073	0.00139	0.19513	0.00290	0.3043	0.0079	4.647	0.140	95	1 812	23
17.1	125	253	59	0.047	0.10903	0.00168	0.58623	0.00494	0.3203	0.0083	4.815	0.153	100	1 783	28
18.1	83	32	47	0.017	0.18529	0.00139	0.10895	0.00152	0.5111	0.0134	13.058	0.367	99	2 701	12
19.1	124	84	43	0.073	0.11013	0.00134	0.19429	0.00275	0.3115	0.0081	4.730	0.142	97	1 802	22
20.1	152	95	88	0.078	0.18148	0.00120	0.17022	0.00194	0.4979	0.0129	12.459	0.342	98	2 666	11
21.1	144	53	56	-0.029	0.12396	0.00105	0.10630	0.00151	0.3646	0.0094	6.232	0.176	100	2 014	15
22.1	178	197	69	0.246	0.10837	0.00109	0.31803	0.00267	0.3157	0.0081	4.718	0.136	100	1 772	18
23.1	136	127	47	0.054	0.10978	0.00118	0.22401	0.00250	0.3011	0.0078	4.557	0.133	94	1 796	20
24.1	217	83	94	0.024	0.14734	0.00091	0.09842	0.00133	0.4042	0.0104	8.210	0.223	95	2 315	11
25.1	90	81	34	0.142	0.10702	0.00169	0.25723	0.00384	0.3229	0.0084	4.765	0.153	103	1 749	29
26.1	464	469	145	0.143	0.20442	0.00095	0.12494	0.00129	0.2724	0.0070	7.679	0.204	54	2 862	8
27.1	165	96	57	0.001	0.10933	0.00095	0.17103	0.00164	0.3117	0.0080	4.699	0.133	98	1 788	16
28.1	206	107	69	0.077	0.11191	0.00097	0.15148	0.00180	0.3072	0.0079	4.739	0.133	94	1 831	16
29.1	150	58	66	0.095	0.14583	0.00111	0.10891	0.00171	0.4086	0.0105	8.215	0.228	96	2 298	13
30.1	72	83	28	0.013	0.10930	0.00140	0.34150	0.00362	0.3162	0.0083	4.765	0.145	99	1 788	23
31.1	66	48	25	-0.099	0.11161	0.00190	0.21439	0.00410	0.3329	0.0087	5.124	0.169	101	1 826	31
32.1	125	137	83	0.032	0.18142	0.00117	0.30649	0.00224	0.5165	0.0134	12.921	0.355	101	2 666	11
33.1	118	70	42	0.090	0.10842	0.00121	0.17198	0.00236	0.3269	0.0085	4.887	0.144	103	1 773	20
34.1	172	113	63	-0.005	0.10988	0.00085	0.19393	0.00158	0.3287	0.0085	4.980	0.138	102	1 797	14
35.1	204	58	66	-0.066	0.11176	0.00120	0.08471	0.00230	0.3132	0.0081	4.826	0.140	96	1 828	20

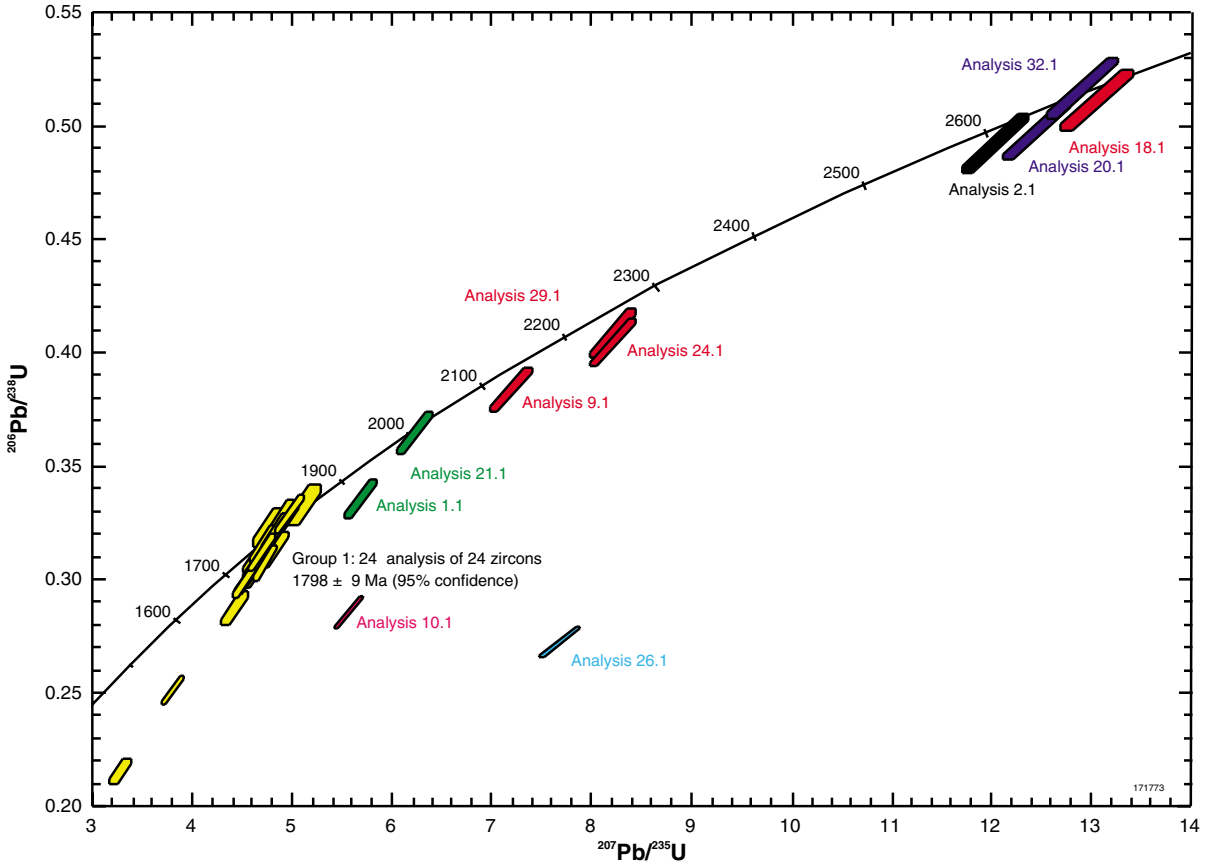

Figure 2. Concordia plot for sample 171773: quartz sandstone, Ingebong Hills

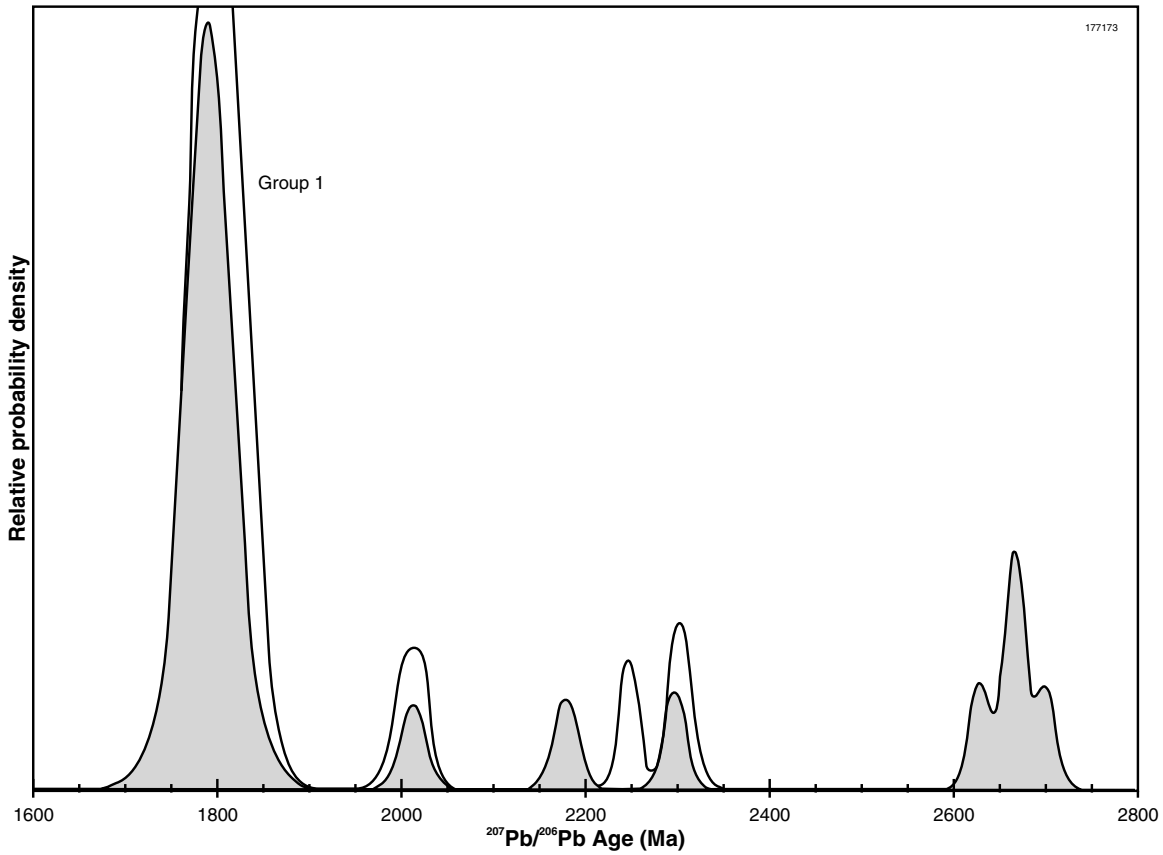

Figure 3. Gaussian-summation probability density plot for sample 171773: quartz sandstone, Ingebong Hills